PH 101

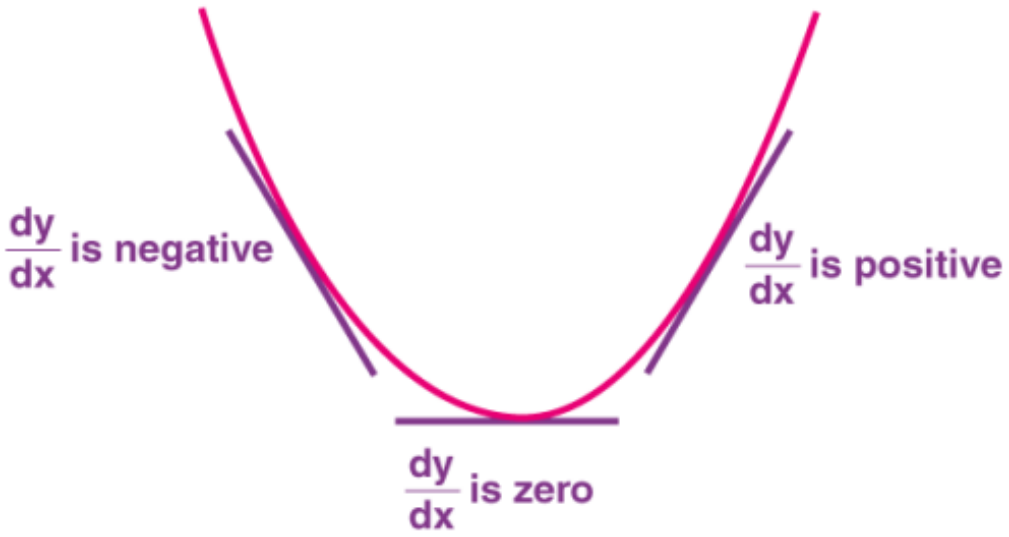
Lecture 02

Mathematical Machinery-1

A Quick look at Single Variable Calculus

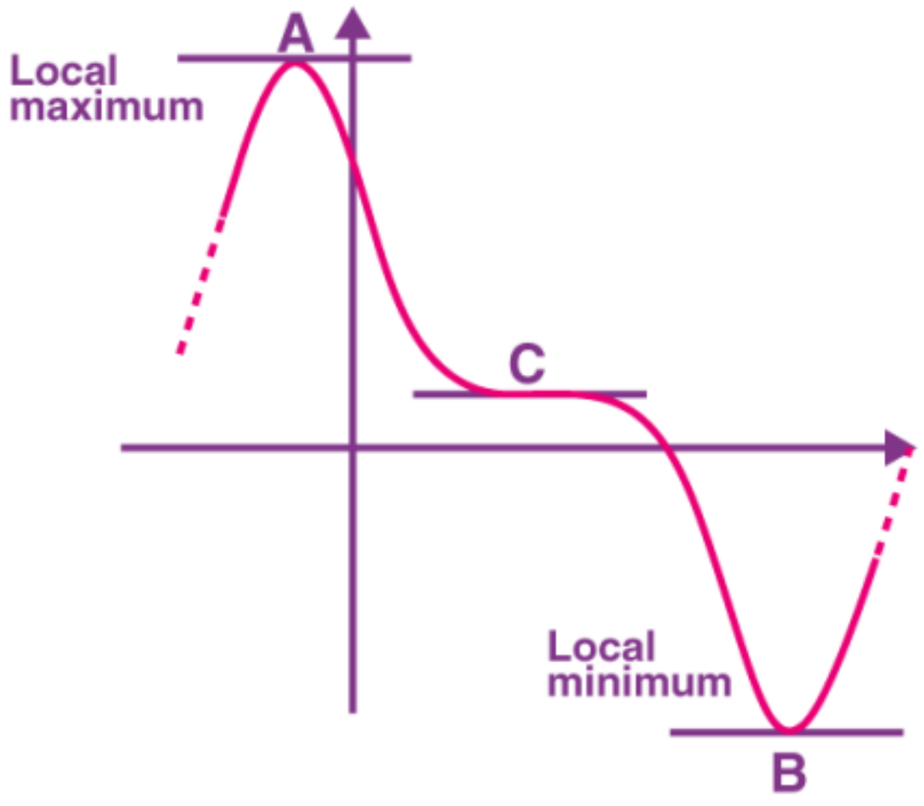
Maxima and Minima of a Function

Maxima and minima in calculus are found by using the concept of derivatives. As we know, the concept of the derivatives gives us information regarding the gradient/slope of the function, we locate the points where the gradient is zero, and these points are called turning points/stationary points. These are points associated with the largest or smallest values (locally) of the function.



The figure for the curve with stationary points is shown. Thus, it can be seen from the figure that before the slope becomes zero, it was negative; after it gets zero, it becomes positive. It can be said **dy/dx** is **-ve** before the stationary point **dy/dx** is **+ve** after the stationary point. Hence, it can be said that **d^2y/dx^2** is positive at the stationary point. Therefore, it can be said wherever the double derivative is positive, it is the point of minima. Vice versa, wherever the double derivative is negative is the point of maxima on the curve. This is also known as the second derivative test.

Stationary points are the points where the slope of the graph becomes zero. In other words, the tangent of the function becomes horizontal, and $dy/dx = 0$. All the stationary points, A, B and C, are given in the figure shown below. And the points in which the function changes its path, if it was going upward; it will go downward and vice versa, i.e., points A and B are turning points since the curve changes its path. But point C is not a turning point, although the graph is flat for a short period of time but continues to go down from left to right.



Some Worked-out examples

Question 1

Find the turning points of the function

$$y = 4x^3 + 8x^2 + 4x + 10$$

Answer

$$\frac{dy}{dx} = 0 \quad \text{for turning points}$$

$$\frac{dy}{dx} = 12x^2 + 16x + 4 = 0$$

$$\Rightarrow \quad 3x^2 + 4x + 1 = 0$$

$$\Rightarrow \quad (x+1)(3x+1) = 0$$

$$\Rightarrow \quad \left.\begin{aligned} x &= -1 \\ x &= -\frac{1}{3} \end{aligned}\right\} \text{ are turning points}$$

Second derivative test

At $x = -1$; $\frac{d^2y}{dx^2} = 24x + 16 \Big|_{x=-1}$

$= -8$

Hence $x = -1$ is a maxima

At $x = -\frac{1}{3}$, $\frac{d^2y}{dx^2} = 24x + 16 \Big|_{x=-\frac{1}{3}}$

$= -8 + 16 = 8$

Hence $x = -\frac{1}{3}$ is a minima

Question 2: Find the local maxima and minima of the function $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$.

Answer:

For stationary points, $f'(x) = 0$.

$f'(x)= 12x^3 + 12x^2 - 24x = 0$

$\Rightarrow 12x(x^2 + x - 2) = 0$

$\Rightarrow 12x(x - 1)(x + 2) = 0$

=> Hence, $x = 0$, $x = 1$ and $x = -2$

Second derivative test:

$f''(x) = 36x^2 + 24x - 24$

$f''(x) = 12(3x^2 + 2x - 2)$

At $x = -2$

$f''(-2) = 12(3(-2)^2 + 2(-2) - 2) = 12\ (12 - 4 - 2) = 12(6) = 72 > 0$

At $x = 0$

$f''(0) = 12(3(0)^2 + 2(0) - 2) = 12(-2) = -24 < 0$

At $x = 1$

$f''(1) = 12(3(1)^2 + 2(1) - 2) = 12(3+2-2) = 12(3) = 36 > 0$

Therefore, by the second derivative test, $x=0$ is the point of local maxima, while $x = -2$ and $x = 1$ are the points of local minima.

A stone is thrown in the air. Its height at any time t is given by $h = -5t + 10t + 4$. Find its maximum height

Solution:

Given $h = -5t^2 + 10t + 4$

$dh/dt = -10t + 10$

Now find when $dh/dt = 0$

$dh/dt = 0 \Rightarrow -10t + 10 = 0$

$\Rightarrow -10t = -10$

$t = 10/10 = 1$

Height at $t = 1$ is given by $h = -5 \times 1^2 + 10 \times 1 + 4$

$= -5 + 10 + 4$

$= 9$

Hence, the maximum height is 9 m.

Partial Derivatives

The calculus of multivariable functions is a straightforward generalization of single-variable calculus.

Multivariable differential calculus revolves around the concept of partial derivatives

Let us consider a function of the variables (x, y, z):

F(x, y, z)

For every value of x, y, z, there is a unique value of F(x, y, z) that we assume varies smoothly as we vary the coordinates.

Say, we are examining the neighborhood of a point x, y, z, and we want to know the rate at which **F** varies as we change x while keeping y and z fixed. We can just imagine that y and z are fixed parameters, so the only variable is x. The derivative of **F** is then defined by:

$$\frac{dF}{dx} = \lim_{\Delta x \to 0} \frac{\Delta F}{\Delta x} \rightarrow (1)$$

where

$$\Delta F = F([x + \Delta x], y, z) - F(x, y, z) \rightarrow (2)$$

$$\frac{dF}{dx} = \lim_{\Delta x \to 0} \frac{\Delta F}{\Delta x} \rightarrow (1)$$

where :

$$\Delta F = F([x+\Delta x], y, z) - F(x, y, z) \rightarrow (2)$$

The derivative defined by Eq. (1) and (2) is called the partial derivative of F with respect to x and is denoted

$$\frac{\partial F}{\partial x}$$

Similarly, we can construct the partial derivative with respect to either of the other variables:

$$\frac{\partial F}{\partial y} = \lim_{\Delta y \to 0} \frac{\Delta F}{\Delta y}$$

Let us think of $\frac{\partial F}{\partial x}$ as itself being a function of x, y, z. Say,

$$G(x,y,z) = \frac{\partial F}{\partial x} \equiv \partial_x F$$

This function G can be differentiated. We can define the second-order partial derivative with respect to x:

$$\frac{\partial G}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial x}\right) = \frac{\partial^2 F}{\partial x^2}$$

$$= \partial_{xx} F$$

Mixed Partial derivatives

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial F}{\partial y}\right) = \partial_{xy} F$$

The mixed derivatives do not depend on the order in which the derivatives are carried out.

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}$$

or $\partial_{xy} F = \partial_{yx} F$

EXAMPLE

Find all the first and second order partial derivatives of the function $z = \sin xy$.

Solution.

$$\begin{aligned}
\frac{\partial z}{\partial x} &= y\cos xy \\
\frac{\partial z}{\partial y} &= x\cos xy \\
\frac{\partial^2 z}{\partial x^2} &= -y^2 \sin xy \\
\frac{\partial^2 z}{\partial y^2} &= -x^2 \sin xy \\
\frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}(x\cos xy) = x(-y\sin xy) + \cos xy = -xy\sin xy + \cos xy \\
\frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}(y\cos xy) = y(-x\sin xy) + \cos xy = -xy\sin xy + \cos xy
\end{aligned}$$

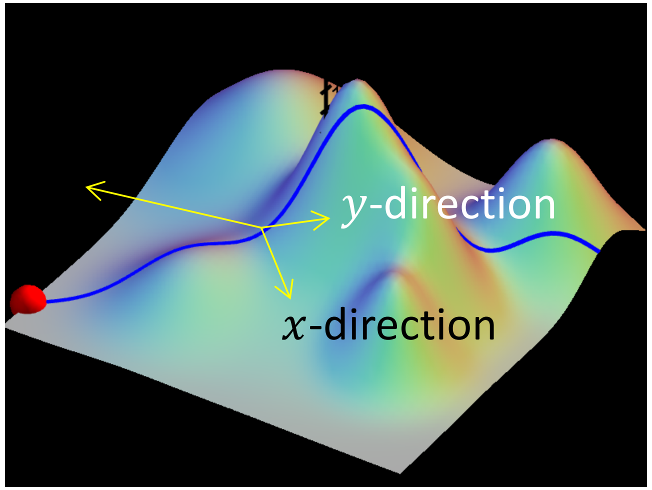
How much the function F(x, y,z) changed when x is increased by dx, keeping y and z fixed?

$$[dF]_{dx} = \left(\frac{\partial F}{\partial x}\right) dx$$

$$= (rate\ of\ change\ in\ x\ direction) \times (amount\ of\ change\ in\ x)$$

How much the function F(x, y,z) changed when x is increased by *dx*, y by *dy* and z by *dz*?

$$dF = \left(\frac{\partial F}{\partial x}\right) dx + \left(\frac{\partial F}{\partial y}\right) dy + \left(\frac{\partial F}{\partial z}\right) dz$$



Generalizing for a function, which depends on several variables $F(q_1, q_2, q_3 \dots q_n)$

$$dF = \left(\frac{\partial F}{\partial q_1}\right) dq_1 + \left(\frac{\partial F}{\partial q_2}\right) dq_2 \;\dots\dots + \left(\frac{\partial F}{\partial q_n}\right) dq_n = \sum_i \left(\frac{\partial F}{\partial q_i}\right) dq_i$$

CHAIN RULE FOR PARTIAL DERIVATIVE

Recall, the chain rule for ordinary derivative first:

If $\quad y = f(u) \quad$ and $\quad u = g(x)$

Then,
$$\frac{dy}{dx} = \frac{dy}{du}\,\frac{du}{dx}$$

u : intermediate variable

x : independent variable

For partial derivatives the chain rule is more complicated. It depends on how many intermediate variables and how many independent variables are present.

(1) if $z = f(x, y)$ and x and y are functions of t ($x = x(t)$ and $y = y(t)$) then z is ultimately a function of t only and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

(2) if $w = f(x, y, z)$ and $x = x(t)$, $y = y(t)$, $z = z(t)$ then w is ultimately a function of t only and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}$$

(3) if $z = f(x, y)$ and $x = x(u, v)$, $y = y(u, v)$ then z is a function of u and v and

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}$$
$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

EXAMPLES

(1) Let $z = x^2y$, $x = t^2$ and $y = t^3$. Calculate dz/dt by (a) the chain rule, (b) expressing z as a function of t and finding dz/dt directly.

Solution. (a) by the chain rule

$$\begin{aligned}
\frac{dz}{dt} &= \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} \\
&= (2xy)(2t) + (x^2)(3t^2) \\
&= 4xyt + 3x^2t^2 \\
&= 4t^2t^3t + 3t^4t^2 \\
&= 7t^6
\end{aligned}$$

(b) $z = x^2y$ and $x = t^2$, $y = t^3$ so $z = t^4t^3 = t^7$. Differentiating gives $dz/dt = 7t^6$.

(2) Let $w = xy + z$ with $x = \cos t$, $y = \sin t$ and $z = t$. Calculate dw/dt.

Solution.

$$\begin{aligned}
\frac{dw}{dt} &= \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt} \\
&= y(-\sin t) + x(\cos t) + (1)(1) \\
&= -\sin^2 t + \cos^2 t + 1
\end{aligned}$$

Gradient operator

Suppose, we have a function of three variables—say, the temperature T(x, y, z) in this room. (Start out in one corner, and set up a system of axes; then for each point (x, y, z) in the room, T gives the temperature at that spot.) We want to generalize the notion of "derivative" to functions like T , which depend not on one but on three variables.

A derivative is supposed to tell us how fast the function varies, if we move a little distance. But this time the situation is more complicated, because it depends on what direction we move!

Fortunately, we now know:

$$dT = \left(\frac{\partial T}{\partial x}\right) dx + \left(\frac{\partial T}{\partial y}\right) dy + \left(\frac{\partial T}{\partial z}\right) dz$$

This tells us how T changes when we alter all three variables by the infinitesimal amounts dx, dy, dz. Notice that we do not require an infinite number of derivatives—three will suffice: the partial derivatives along each of the three coordinate directions

$$dT = \left(\frac{\partial T}{\partial x}\right) dx + \left(\frac{\partial T}{\partial y}\right) dy + \left(\frac{\partial T}{\partial z}\right) dz$$

$$\begin{aligned} dT &= \left(\frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}\right) \cdot (dx\,\hat{\mathbf{x}} + dy\,\hat{\mathbf{y}} + dz\,\hat{\mathbf{z}}) \\ &= (\nabla T) \cdot (d\mathbf{l}), \end{aligned}$$

Where $\nabla T \equiv \frac{\partial T}{\partial x}\hat{\mathbf{x}} + \frac{\partial T}{\partial y}\hat{\mathbf{y}} + \frac{\partial T}{\partial z}\hat{\mathbf{z}}$ is called the gradient of T

$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}$ is called the **del** operator

Geometrical Interpretation of the Gradient:

Like any vector, the gradient has magnitude and direction. To determine its geometrical meaning, let's rewrite the dot product

$$dT = \nabla T \cdot d\mathbf{l} = |\nabla T||d\mathbf{l}|\cos\theta$$

where θ is the angle between ∇T and $d\mathbf{l}$. Now, if we fix the magnitude $|d\mathbf{l}|$ and search around in various directions (that is, vary θ), the maximum change in T occurs when $\theta = 0$ (for then $\cos\theta = 1$). That is, for a fixed distance $|d\mathbf{l}|$, dT is greatest when we move in the same direction as ∇T. Thus:

The gradient ∇T points in the direction of maximum increase of the function T

The magnitude $|\nabla T|$ gives the slope (rate of increase) along this maximal direction

Example:

The height of a certain hill (in meter) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$

where y is the distance (in km) north, x the distance east of IIT Guwahati

(a) Where is the top of the hill located?
(b) How high is the hill?
(c) How steep is the slope (in meter per kilometer) at a point 1 km north and one km east of IIT Guwahati? In what direction is the slope steepest, at that point?

Solution:

(a) $\nabla h = 10[(2y - 6x - 18)\,\hat{\mathbf{x}} + (2x - 8y + 28)\,\hat{\mathbf{y}}]$. $\nabla h = 0$ at summit, so

$$\left.\begin{array}{l} 2y - 6x - 18 = 0 \\ 2x - 8y + 28 = 0 \Longrightarrow 6x - 24y + 84 = 0 \end{array}\right\} 2y - 18 - 24y + 84 = 0.$$

$$22y = 66 \Longrightarrow y = 3 \Longrightarrow 2x - 24 + 28 = 0 \Longrightarrow x = -2.$$

Top is 3 km north, 2 km west, of IIT Guwahati

(b)

Putting in $x = -2$, $y = 3$:

$$h = 10(-12 - 12 - 36 + 36 + 84 + 12) = \boxed{720} \text{ meter}$$

(c)

Putting in $x = 1$, $y = 1$: $\nabla h = 10[(2 - 6 - 18)\,\hat{\mathbf{x}} + (2 - 8 + 28)\,\hat{\mathbf{y}}] = 10(-22\,\hat{\mathbf{x}} + 22\,\hat{\mathbf{y}}) = 220(-\hat{\mathbf{x}} + \hat{\mathbf{y}})$.

$|\nabla h| = 220\sqrt{2}$ m/km   Direction: northwest